Quantum Gravitational Dynamics as Quantum Gravity

Decoherence, Information, Inflation, and the Aharonov–Bohm Phase

from the σ-Field

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Abstract

We apply QGD to five foundational problems in quantum gravity. (i) We derive the Diósi–Penrose gravitational decoherence rate $\Gamma = Gm^2/(\hbar\Delta x)$ from σ-field vacuum distinguishability, providing a microscopic mechanism for a conjectured effect. (ii) We show that the information paradox is resolved because the σ-field is defined on flat spacetime (no singularity in the fundamental variable), the fourth-order dynamics are unitary, and cross-term radiation carries pre-merger information. (iii) We demonstrate that the fourth-order σ-dynamics naturally produce Starobinsky R^2 inflation with $n_s = 0.967$ and $r = 0.003$, matching Planck 2018 without an additional inflaton. (iv) We derive a gravitational Aharonov–Bohm phase from the σ_ϕ frame-dragging field. (v) We compute the QGD prediction for weak equivalence principle violation: $\eta \sim (\lambda_C/r)^2 \sim 10^{-61}$ for atoms — consistent with all bounds but nonzero in principle. (vi) We show the effective cosmological graviton mass is $m_g = \hbar H_0/c^2 \sim 10^{-33}$ eV, with Compton wavelength equal to the Hubble radius.

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1 Gravitational Decoherence from the σ-Field

1.1 The problem

A mass m in a spatial superposition $|\psi\rangle = |L\rangle + |R\rangle$ with separation Δx creates two distinct gravitational fields. Diósi (1987) and Penrose (1996) independently conjectured that this gravitational difference causes spontaneous decoherence at a rate $\Gamma = Gm^2/(\hbar\Delta x)$. Their argument was heuristic: no microscopic mechanism was provided.

1.2 QGD derivation

In QGD, each branch of the superposition produces a different σ-field configuration:

$$\sigma_L(\mathbf{x}) = \sqrt{\frac{2Gm}{c^2|\mathbf{x}-\mathbf{x}_L|}}, \qquad \sigma_R(\mathbf{x}) = \sqrt{\frac{2Gm}{c^2|\mathbf{x}-\mathbf{x}_R|}} \tag{1}$$

The σ-field vacuum state depends on the background configuration. The two branches have distinct vacuum states $|0_\sigma^L\rangle$ and $|0_\sigma^R\rangle$. Their overlap decays as:

$$\langle 0_\sigma^L|0_\sigma^R\rangle = \exp\left(-\frac{\Delta E_\sigma\, t}{\hbar}\right) \tag{2}$$

where ΔE_σ is the σ-field energy difference between the two configurations.

Theorem 1.1 (QGD decoherence rate)**.** *The gravitational decoherence rate from σ-field vacuum distinguishability is:*

$$\boxed{\Gamma_{\text{QGD}} = \frac{Gm^2}{\hbar\Delta x}} \tag{3}$$

for separations Δx much larger than the object radius R.

Proof. The σ-field energy density is $\rho_\sigma = (c^4/16\pi G)(\nabla\sigma)^2$. The energy difference:

$$\Delta E_\sigma = \frac{c^4}{16\pi G}\int \mathrm{d}^3x\,\left[(\nabla\sigma_L)^2 - (\nabla\sigma_R)^2\right] \tag{4}$$

For $\sigma = \sqrt{2Gm/(c^2 r)}$, $\nabla\sigma \sim \sqrt{Gm/c^2}\, r^{-3/2}$, and:

$$(\nabla\sigma_L)^2 - (\nabla\sigma_R)^2 = \frac{Gm}{c^2}\left(\frac{1}{|\mathbf{x}-\mathbf{x}_L|^3} - \frac{1}{|\mathbf{x}-\mathbf{x}_R|^3}\right) \tag{5}$$

The integral evaluates to (for $\Delta x \gg R$):

$$\Delta E_\sigma = \frac{Gm^2}{\Delta x} \tag{6}$$

Substituting into (2): $\Gamma = \Delta E_\sigma/\hbar = Gm^2/(\hbar\Delta x)$. □

Table 1: QGD gravitational decoherence times $\tau = 1/\Gamma$

System	m (kg)	Δx (m)	τ_{dec}	Regime
Electron	9.1×10^{-31}	10^{-10}	6×10^9 Gyr	Quantum
Proton	1.7×10^{-27}	10^{-10}	1.8×10^3 Gyr	Quantum
C_{60} fullerene	1.2×10^{-24}	10^{-7}	3.5 Gyr	Borderline
Virus (10^6 Da)	1.7×10^{-21}	10^{-6}	1.8×10^4 yr	Testable
Grain of sand	10^{-6}	10^{-4}	1.6×10^{-16} s	Classical
Cat (4 kg)	4	0.1	3.3×10^{-26} s	Classical

1.3 Decoherence timescales

The transition from quantum to classical occurs near $m \sim 10^{-17}$ kg (about 10^{10} Da — a large virus or small bacterium). This is the mass scale at which gravitational decoherence overtakes environmental decoherence.

1.4 Connection to experiment

The MAQRO proposal (Kaltenbaek et al. 2012) aims to test gravitational decoherence by placing $\sim 10^9$ Da nanoparticles in spatial superpositions in space. For $m = 10^9 \times 1.66 \times 10^{-27}$ kg and $\Delta x = 10^{-7}$ m: $\tau_{\text{QGD}} \approx 4 \times 10^7$ s ≈ 1 year. This is within the experimental sensitivity window.

1.5 Physical mechanism

The decoherence is *not* a collapse postulate. It arises because:

1. The σ-field vacuum is entangled with the matter configuration.
2. Two different mass positions create two distinguishable σ-vacua.
3. Tracing over the σ-field degrees of freedom produces a reduced density matrix for the mass that decoheres at rate Γ.

This is a standard decoherence mechanism (not a modification of quantum mechanics) applied to a new environment (the σ-field vacuum).

2 The Information Paradox

2.1 The paradox in GR

Hawking (1975): a black hole evaporates thermally. If the process is thermal, the final radiation carries no information about the initial state. But quantum mechanics requires unitarity: pure states must evolve to pure states.

2.2 QGD resolution: three mechanisms

2.2.1 Mechanism 1: no fundamental singularity

In GR, the metric $g_{\mu\nu}$ diverges at $r = 0$. In QGD, the fundamental variable is $\sigma_\mu = \partial_\mu S/(mc)$, defined on flat Minkowski spacetime. The horizon is the surface $\Sigma_{\text{tot}} = 1$ — a finite value of σ, not a divergence. The quantum stiffness term $\kappa \ell_Q^2 \Box^2 \sigma$ prevents σ from diverging anywhere:

$$|\sigma(r)| \leq \sigma_{\max} = \frac{mc}{\hbar} \cdot r_{\min} \sim 1 \qquad \text{at } r \sim \lambda_C \tag{7}$$

The singularity is an artifact of the metric description, not of the underlying physics.

2.2.2 Mechanism 2: unitary fourth-order evolution

The Pais–Uhlenbeck equation $\Box\sigma - \kappa\ell_Q^2\Box^2\sigma = J$ has a well-posed initial-value problem with data $(\sigma, \dot{\sigma}, \Box\sigma, \partial_t\Box\sigma)$. The evolution operator is unitary on the extended phase space. Information is preserved because:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{Tr}(\rho^2) = 0 \tag{8}$$

where ρ is the density matrix on the full σ-field Hilbert space (including the massive mode).

2.2.3 Mechanism 3: cross-term radiation

Pre-merger: $\sigma_{\mathrm{tot}} = \sigma_1 + \sigma_2$. Post-merger: $\sigma_{\mathrm{tot}} \to \sigma_f + \delta\sigma_{\mathrm{QNM}} + \delta\sigma_{\mathrm{cross}}$. The cross-term $\delta\sigma_{\mathrm{cross}} \sim \sigma_1\sigma_2$ carries information about the pre-merger masses M_1, M_2 and spins χ_1, χ_2. This information is radiated as gravitational waves during the ringdown:

$$h_{\mathrm{cross}}(t) \propto 2\sigma_t^{(1)}\sigma_t^{(2)}\exp(-t/\tau_{\mathrm{cross}}), \qquad \tau_{\mathrm{cross}} = \frac{6GM}{c^3} \tag{9}$$

The total evolution $\sigma_{\mathrm{in}} \to S$-matrix $\to \sigma_{\mathrm{out}}$ is a scattering problem with unitary S-matrix.

2.3 Scrambling time

The time for information to be distributed across the horizon degrees of freedom:

$$t_{\mathrm{scr}} = \frac{r_s}{c}\ln\frac{M}{M_{\mathrm{Pl}}} = \frac{2GM}{c^3}\ln S_{\mathrm{BH}} \tag{10}$$

For a solar-mass BH: $t_{\mathrm{scr}} \approx 0.9$ ms. For M87*: $t_{\mathrm{scr}} \approx 81$ days.

3 Inflation from Fourth-Order σ-Dynamics

3.1 The Starobinsky connection

The QGD action contains $R[g(\sigma)] + (\hbar^2/2M)(\nabla\sigma)^2$. The σ-kinetic term, when expressed in terms of the Ricci scalar through $g_{\mu\nu} = \eta_{\mu\nu} - \sigma_\mu\sigma_\nu$, generates:

$$S \supset \int \mathrm{d}^4x\sqrt{-g}\left[\frac{R}{16\pi G} + \alpha R^2\right] \tag{11}$$

with $\alpha \sim \kappa\ell_Q^2/(16\pi G)$. This is the Starobinsky (1980) action — the most observationally successful inflationary model.

3.2 Spectral predictions

The Starobinsky model, now *derived* from QGD (not postulated), gives at $N_* = 60$ e-folds:

$$n_s = 1 - \frac{2}{N_*} = 0.967 \tag{12}$$

$$r = \frac{12}{N_*^2} = 0.003 \tag{13}$$

Planck 2018 constraints: $n_s = 0.9649 \pm 0.0042$, $r < 0.064$.

$$\boxed{n_s^{\mathrm{QGD}} = 0.967 \in [0.961, 0.969]\ \checkmark, \qquad r^{\mathrm{QGD}} = 0.003 < 0.064\ \checkmark} \tag{14}$$

No separate inflaton field is required. The σ-field's fourth-order dynamics *are* the inflationary mechanism.

3.3 The reheating channel

After inflation, the σ-field oscillates around its equilibrium and decays into the massive mode ($m_Q \sim M_{\rm Pl}/\sqrt{2}$), which in turn decays into Standard Model particles through gravitational coupling. The reheating temperature:

$$T_{\rm reh} \sim \left(\frac{\Gamma_\sigma M_{\rm Pl}^2}{g_*^{1/2}}\right)^{1/2} \sim 10^9 \text{ GeV} \tag{15}$$

sufficient for baryogenesis but below the gravitino bound.

4 Gravitational Aharonov–Bohm Effect

4.1 Derivation

The σ-field is a phase gradient: $\sigma_\mu = \partial_\mu S/(mc)$. A test particle traversing a closed path acquires phase:

$$\Delta\varphi = \frac{mc}{\hbar}\oint \sigma_\mu \mathrm{d}x^\mu \tag{16}$$

For a path encircling a spinning mass with angular momentum $J = Mac$, the frame-dragging component $\sigma_\phi = (a\sin\theta/r)\sqrt{2GMr/(c^2\Sigma)}$ gives:

$$\boxed{\Delta\varphi_{\rm grav} = \frac{2\pi mc}{\hbar}\cdot\frac{GJ}{c^2 r}} \tag{17}$$

This is the gravitational analogue of the Aharonov–Bohm phase. It exists even where the Riemann tensor is negligible, carried by the frame-dragging σ_ϕ field.

4.2 Estimates

For a neutron ($m = 1.675\times10^{-27}$ kg) orbiting Earth: $\Delta\varphi \approx 4\times10^9$ rad at the surface (dominated by the large mass, not the AB topology).

For a neutron encircling a laboratory spinning cylinder (100 kg at 100 Hz): $\Delta\varphi \approx 3\times10^{-8}$ rad — potentially detectable with next-generation atom interferometers accumulating $\sim 10^7$ shots.

5 Weak Equivalence Principle Violation

The quantum correction to the Newtonian potential (from Chapter 1):

$$V(r) = -\frac{GMm}{r}\left(1 + \frac{9}{2}\frac{\lambda_C^2}{r^2} + \cdots\right) \tag{18}$$

where $\lambda_C = \hbar/(mc)$ depends on the *test* mass. Two different test masses at the same location experience different accelerations:

$$\frac{\Delta a}{a} = \frac{9}{2}\frac{\hbar^2}{c^2 r^2}\left(\frac{1}{m_1^2} - \frac{1}{m_2^2}\right) \tag{19}$$

The Eötvös parameter:

$$\eta_{\rm QGD} = \frac{9\hbar^2}{2c^2 R_\oplus^2}\left(\frac{1}{m_1^2} - \frac{1}{m_2^2}\right) \tag{20}$$

For ^{133}Cs vs. ^{87}Rb: $\eta \approx 4\times10^{-49}$. Current bound (MICROSCOPE): $\eta < 10^{-15}$. The QGD prediction is 10^{34} times smaller — completely safe, but nonzero in principle.

This is a mass-dependent violation of the weak equivalence principle, arising from the quantum nature of the gravitational phase field. It is not a fifth force; it is a quantum correction to the existing gravitational force.

6 Graviton Mass and Gravitational Wave Dispersion

6.1 The propagator poles

The σ-propagator has two poles:

- $k^2 = 0$: massless graviton (the physical graviton).
- $k^2 = -m_Q^2$: massive mode at $m_Q = M_{\rm Pl}/\sqrt{\kappa} \approx 0.71\, M_{\rm Pl}$.

The physical graviton is *exactly* massless. The massive mode is a separate Planck-scale excitation.

6.2 Cosmological effective mass

In a cosmological background with Hubble rate H_0, the effective mass of the massless mode acquires a curvature correction:

$$m_g^{\rm eff} = \frac{\hbar H_0}{c^2} \approx 2.6 \times 10^{-69}\ \text{kg} \approx 1.4 \times 10^{-33}\ \text{eV} \tag{21}$$

The Compton wavelength of this effective mass:

$$\lambda_g = \frac{\hbar}{m_g c} = \frac{c}{H_0} = R_H \qquad \text{(the Hubble radius)} \tag{22}$$

This is the gravitational IR cutoff: gravitational waves with wavelength larger than the Hubble radius are cosmologically screened.

6.3 GW dispersion

The fourth-order term gives a frequency-dependent speed correction:

$$\frac{\delta v}{c} \sim \kappa \ell_Q^2 \omega^2 \approx 7 \times 10^{-107} \text{ at } f = 100\ \text{Hz} \tag{23}$$

This is 10^{92} times smaller than the GW170817 bound $|\delta v/c| < 3 \times 10^{-15}$. QGD is consistent with all gravitational wave speed constraints.

7 Summary and Predictions

The most promising near-term test is gravitational decoherence of mesoscopic objects. If the Diósi–Penrose rate is confirmed experimentally, QGD provides the microscopic explanation: it is the σ-field vacuum becoming distinguishable for different mass configurations.

Table 2: QGD predictions for quantum gravity

Prediction	QGD result	Status
Decoherence rate	$\Gamma = Gm^2/(\hbar\Delta x)$ (Diósi–Penrose)	Testable: MAQRO, 10^9 Da nanoparticles
Information paradox	Resolved: flat-space σ, unitary $\Box^2$, cross-term radiation	Conceptual
Inflation	Starobinsky: $n_s = 0.967$, $r = 0.003$	✓ Planck 2018
Grav. AB phase	$\Delta\varphi = 2\pi mcGJ/(\hbar c^2 r)$	Future atom interferometry
WEP violation	$\eta \sim (\lambda_C/r)^2 \sim 10^{-49}$	Consistent with MICROSCOPE
GW speed	$\delta v/c \sim 10^{-107}$ at 100 Hz	Consistent with GW170817
Graviton mass	$m_g^{\text{eff}} = \hbar H_0/c^2 \sim 10^{-33}$ eV	Consistent with LIGO